

MongoDB Interview Questions (45 Questions)

BASIC MongoDB QUESTIONS (Level 1)

1. What is MongoDB? How is it different from RDBMS?
 2. What is a document in MongoDB?
 3. What is a collection?
 4. What is BSON? Why does MongoDB use BSON instead of JSON?
 5. What are the advantages of MongoDB?
 6. What is _id field? Can we customize it?
 7. Difference between find() and findOne()
 8. What is the difference between insertOne() and insertMany()?
 9. What are embedded documents?
 10. What is the difference between updateOne(), updateMany() and replaceOne()?
 11. What is schema-less design? Is MongoDB truly schema-less?
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INTERMEDIATE MongoDB QUESTIONS (Level 2)

12. What are indexes in MongoDB? Why are they needed?
 13. Types of indexes in MongoDB.
 14. What is a compound index?
 15. What is the difference between createIndex() and ensureIndex()?
 16. What is aggregation framework?
 17. Difference between aggregate() and find()
 18. Explain \$match, \$group, \$project, \$sort stages.
 19. What is \$lookup? How is it similar to SQL JOIN?
 20. What is the difference between embedding and referencing?
 21. What is ObjectId? How is it generated?
 22. What is explain()? Why is it used?
 23. What is a capped collection?
 24. What are TTL indexes?
 25. Difference between deleteOne() and deleteMany()
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ADVANCED MongoDB QUESTIONS (Level 3)

26. What is replication in MongoDB?
 27. Explain Replica Set architecture.
 28. What is sharding? Why is it required?
 29. What is a shard key? How do you choose it?
 30. What is write concern and read preference?
 31. What is MongoDB transaction?
 32. How does MongoDB ensure ACID properties?
 33. What is journaling in MongoDB?
 34. How do you handle schema validation in MongoDB?
 35. Difference between MongoDB transactions and RDBMS transactions?
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SCENARIO-BASED MONGODB QUESTIONS (Very Important)

36. **E-commerce Scenario**
Orders grow rapidly.
Would you embed order items or reference them? Why?
37. **Performance Scenario**
Query becomes slow after data crosses 10 million records.
How will you optimize it?
38. **Analytics Scenario**
Need daily sales reports.
Which MongoDB feature will you use?
39. **Microservices Scenario**
Multiple services accessing same collection.
How do you ensure data consistency?
40. **High Availability Scenario**
Primary node goes down.
What happens next?
41. **Geo-location Scenario**
Store user location & find nearby users.
How will you design this?
42. **Login System Scenario**
Store login attempts & auto-delete after 24 hrs.
How will you implement this?

43. Migration Scenario

Need to migrate MySQL data to MongoDB.

What challenges will you face?

44. Big Document Scenario

Document size is increasing continuously.

What problems can occur and how will you solve them?

45. Reporting Tool Scenario (Power BI)

MongoDB feeds dashboards every few minutes.

How will you optimize aggregation queries?

Redis Interview Questions (10 Questions)

1. What is Redis?
 2. Why is Redis called an in-memory database?
 3. What data structures does Redis support?
 4. Difference between Redis and Memcached?
 5. What is Redis persistence? Explain RDB and AOF.
 6. What is Redis TTL? How does key expiration work?
 7. What is Redis Pub/Sub?
 8. How is Redis used for caching in real applications?
 9. What is Redis clustering?
 10. How does Redis handle high availability and failover?
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Frequently Asked in System Design

- When to use **MongoDB vs MySQL**
- MongoDB + Redis in same architecture
- Cache-aside vs Write-through caching
- Handling cache invalidation